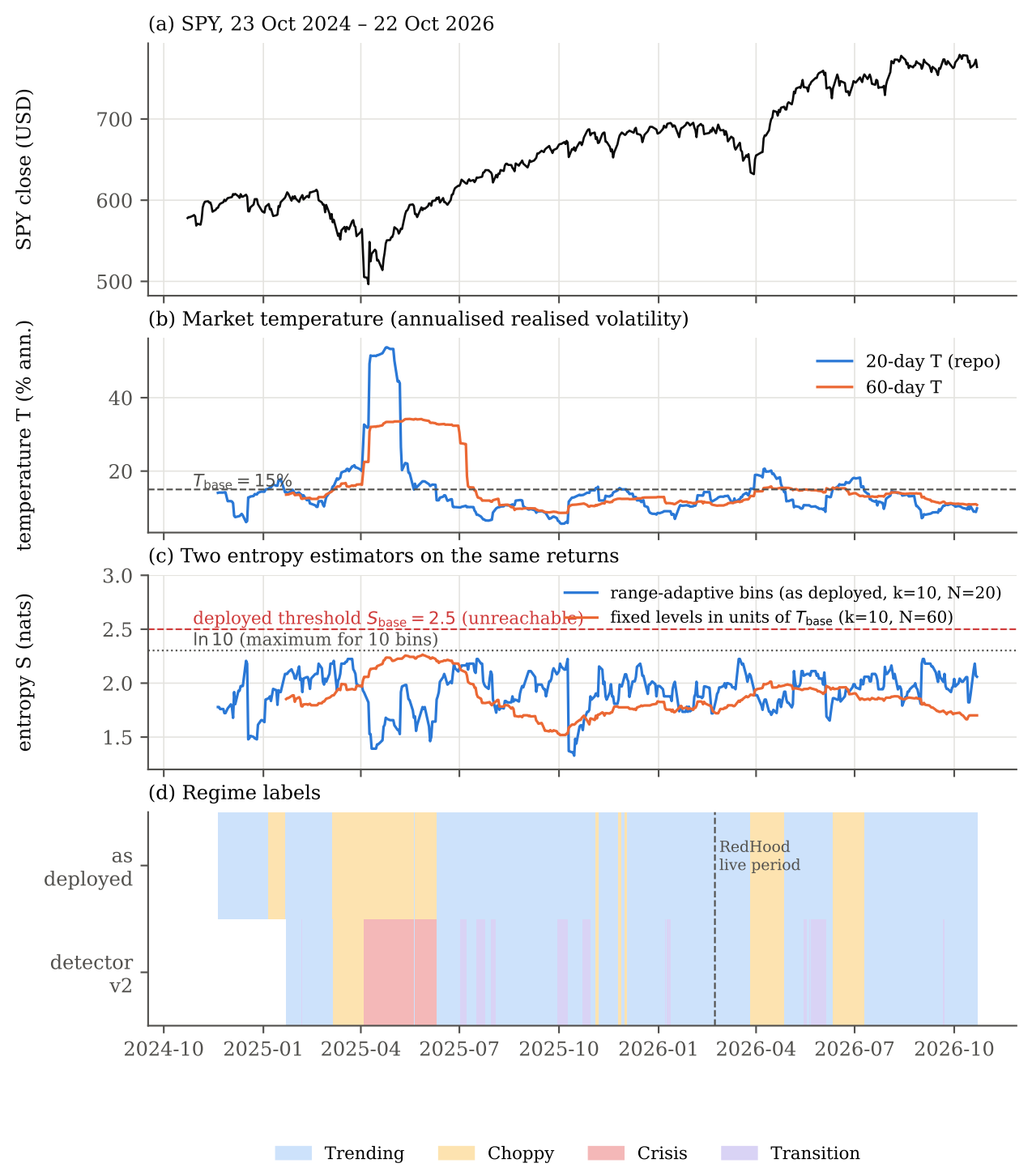
(a) SPY, 23 Oct 2024 – 22 Oct 2026
SPY close (USD)
700
600
500
(b) Market temperature (annualised realised volatility)
temperature T (% ann.)
40
20
20-day T (repo)
60-day T
T_{base} = 15%
(c) Two entropy estimators on the same returns
entropy S (nats)
3.0
2.5
2.0
1.5
range-adaptive bins (as deployed, k=10, N=20)
fixed levels in units of T_{base} (k=10, N=60)
deployed threshold S_{base} = 2.5 (unreachable)
ln 10 (maximum for 10 bins)
(d) Regime labels
as deployed
detector v2
RedHood live period
2024-10
2025-01
2025-04
2025-07
2025-10
2026-01
2026-04
2026-07
2026-10
Trending
Choppy
Crisis
Transition